

CS509 Laboratory Repository

Repository Overview

This repository contains individual assignments for CS509 - Parallel Computer Architecture. Assignment 01 implements General Matrix Multiplication (GEMM) using both simple and blocking algorithms.

Student Details

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- **Program:** M.Tech Artificial Intelligence, IIT Ropar

Language and Environment

- **Language:** C++17
- **Compiler:** MinGW g++ (GCC)
- **Build Tool:** mingw32-make
- **Timing Method:** `std::chrono::high_resolution_clock`, reported in milliseconds (ms)
- **Platform:** Windows 10/11

Directory Structure

```
CS_509_2026AIM1023/  
|-- README.md
```

```

|-- common_wrapper/
|   |-- wrapper.cpp
|   `-- Makefile
|-- assignment_01/
|   |-- driver/
|   |   |-- driver.cpp
|   |-- include/
|   |   |-- gemm.h
|   |   |-- matrix.h
|   |   |-- timer.h
|   |-- src/
|   |   |-- gemm_blocking.cpp
|   |   |-- gemm_simple.cpp
|   |   |-- matrix.cpp
|   |   |-- timer.cpp
|   |-- tests/
|   |   |-- gemm_test_01.txt
|   |   |-- gemm_test_02.txt
|   |   |-- gemm_test_03.txt
|   |   |-- gemm_test_04.txt
|   |   |-- gemm_test_05.txt
|   |   |-- gemm_test_06.txt
|   `-- Makefile

```

Common Wrapper: Build and Usage

The wrapper acts as a main gateway program to compile and run assignments.

```

cd common_wrapper
mingw32-make
.\wrapper.exe

```

The wrapper provides:

- Assignment selection menu
- Compile Assignment 01
- Run one selected test file
- Run all test files
- Clear error messages for missing executables or test files

General Conventions

- One test case per `.txt` file in `assignment_01/tests/`
 - Timing includes only core algorithm execution (excluding file I/O or output printing)
 - Variables use type-prefixed snake_case naming style.
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Assignment 01 - GEMM

Assignment Mode

Single (Individual)

Objective

Implement and compare simple and blocking GEMM algorithms on the same matrix inputs.

Algorithm / Approach

- **Simple GEMM:** Baseline nested triple loop matrix multiplication. It iterates row-by-row (i, j, k) to compute dot products.
- **Blocking GEMM:** Tiled multiplication. Divides matrices into submatrix blocks of size `block_size` to optimize cache hit rates by reuse.

Input Format

```
M K N
A row 0 values
...
A row M-1 values
B row 0 values
...
B row K-1 values
```

File Structure

File	Purpose
-----	-----
driver/driver.cpp	Command-line dispatch, reads matrix inputs, calls timers, outputs correctness / matches.
src/gemm_simple.cpp	Standard triple loop matrix multiplication.
src/gemm_blocking.cpp	Blocked matrix multiplication.
src/matrix.cpp	Matrix parsing, creation, and printing helpers.
src/timer.cpp	High-resolution clock utilities.

Compilation

```
cd assignment_01
mingw32-make
```

Execution

```
# Run a single algorithm on a test file
.\gemm.exe simple tests/gemm_test_02.txt
.\gemm.exe blocking tests/gemm_test_02.txt 32
.\gemm.exe both tests/gemm_test_02.txt 32

# Run all tests in the tests folder
.\gemm.exe --all-tests both 32
```

Test Cases and Result Table

Test File	Input Type / Size	Expected Output	Actual Output	Simple Time (ms)	Blocking Time (ms)	Block Size	Status
-----	----- -----	----- -----	----- -----	----- -----	----- -----	----- -----	----- --
gemm_test_01.txt	A(20x10), B(10x30)	Computed 20x30 matrix	Same as simple	0.000 ms	0.000 ms	32	Pass
gemm_test_02.txt	A(2x3), B(3x2)	[[58,64], [139,154]]	Same as simple	0.000 ms	0.000 ms	32	Pass
gemm_test_03.txt	A(50x50), B(50x50)	Computed 50x50 matrix	Same as simple	1.336 ms	1.336 ms	32	Pass
gemm_test_04.txt	A(100x100), B(100x100)	Computed 100x100 matrix	Same as simple	8.968 ms	8.968 ms	32	Pass
gemm_test_05.txt	A(250x250), B(250x250)	Computed 250x250 matrix	Same as simple	150.12 ms	152.82 ms	32	Pass
gemm_test_06.txt	A(500x500), B(500x500)	Computed 500x500	Same as	1255.03 ms	1269.85 ms	32	Pass

Test File	Input Type / Size	Expected Output	Actual Output	Simple Time (ms)	Blocking Time (ms)	Block Size	Status
		matrix	simple				

Complexity

- **Simple GEMM:** Time complexity $O(MKN)$, Space complexity $O(MK + KN + MN)$
- **Blocking GEMM:** Time complexity $O(MKN)$, Space complexity $O(MK + KN + MN)$

References

- CS509 Assignment 1 Specification
- CS509 Lab Work Guidelines